

Linear Algebra (Summer 2026)

Week 5 Discussion

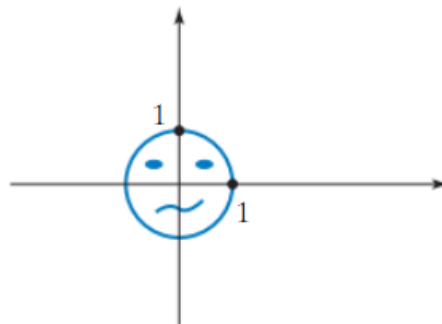
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Problem 1

- a) Let A be a 6×5 matrix. What must a and b be in order to define $T: \mathbb{R}^a \rightarrow \mathbb{R}^b$ by $T(x) = Ax$?
- b) How many rows and columns must a matrix A have in order to define a mapping from $\mathbb{R}^4$ into $\mathbb{R}^5$ by the rule $T(x) = Ax$?

Problem 2

Consider the circular face in the accompanying figure. For each of the following matrices A , draw a sketch showing the effect of the linear transformation $T(x) = Ax$ on this face.



a)

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

b)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Problem 3

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps $u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ into $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and maps $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ into $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$. Use the fact that T is linear to find the images under T of $3u$, $2v$, and $3u + 2v$.

Problem 4

Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $y_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, and $y_2 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$, and let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps e_1 into y_1 and maps e_2 into y_2 . Find the images of $\begin{bmatrix} 5 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

Problem 5

Show that the transformation T defined by $T(x_1, x_2) = (2x_1 - 3x_2, x_1 + 4, 5x_2)$ is not linear.

Problem 6

$$A = \begin{bmatrix} 4 & -2 & 5 & -5 \\ -9 & 7 & -8 & 0 \\ -6 & 4 & 5 & 3 \\ 5 & -3 & 8 & -4 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 7 \\ 5 \\ 9 \\ 7 \end{bmatrix}.$$

[P] Is b in the range of the transformation $x \mapsto Ax$? If so, find an x whose image under T is b .

Problem 7

Assume T is a linear transformation. Find the standard matrix of T .

- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$, $T(e_1) = (3, 1, 3, 1)$ and $T(e_2) = (-5, 2, 0, 0)$, where $e_1 = (1, 0)$ and $e_2 = (0, 1)$.
- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates points about the origin through $3\pi/2$ radians.
- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vertical shear transformation that maps e_1 into $e_1 - 2e_2$ but leaves e_2 unchanged.
- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first reflects points through the vertical x_2 -axis and then rotates points $\pi/2$ radians.

Problem 8

$$T(x_1, x_2, x_3, x_4) = (0, x_1 + x_2, x_2 + x_3, x_3 + x_4)$$

- Find the standard matrix of T .
- Is T one-to-one? Onto?

Problem 9

$$T(x_1, x_2, x_3) = (x_1 - 5x_2 + 4x_3, x_2 - 6x_3)$$

- Find the standard matrix of T .
- Is T one-to-one? Onto?

Problem 10

[P] Let T be the linear transformation whose standard matrix is given. Is T one-to-one? Onto?

a)

$$\begin{bmatrix} 4 & -7 & 3 & 7 & 5 \\ 6 & -8 & 5 & 12 & -8 \\ -7 & 10 & -8 & -9 & 14 \\ 3 & -5 & 4 & 2 & -6 \\ -5 & 6 & -6 & -7 & 3 \end{bmatrix}$$

b)

$$\begin{bmatrix} 9 & 13 & 5 & 6 & -1 \\ 14 & 15 & -7 & -6 & 4 \\ -8 & -9 & 12 & -5 & -9 \\ -5 & -6 & -8 & 9 & 8 \\ 13 & 14 & 15 & 2 & 11 \end{bmatrix}$$